Induction Heads and Model Collapse: How the Real-to-Synthetic Data Ratio Shapes In-Context Learning Circuits

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Abstract

Roughly 20 lines. It should follow the format of the paper where each section is summarised as a sentence (not a fixed rule but a bit of a heuristic).

Keywords: 3 Words or phrases that give the main categories or topics of the work

Introduction

Large language models (LLMs) are trained on internet data increasingly generated by other models. Shumailov et al. (2024) showed that recursive training on such data causes *model collapse*, in which successive generations lose information about the true data distribution, beginning with its tails. In practice, real and synthetic data tend to accumulate, and Gerstgrasser et al. (2024) found that retaining real data avoids collapse. However, both studies measure collapse through model outputs. If recursive training damages a model's internal mechanisms before its outputs degrade, output metrics would understate collapse.

In-context learning (ICL) lets models adapt to new tasks from prompt examples without parameter updates (Olsson et al., 2022; Singh et al., 2024), making its robustness to collapse important for future LLMs. In the task of Singh et al. (2024), a model outputs the label paired with a query symbol, with labels reassigned per sequence so that only the context holds the answer. The task is solved by an induction circuit: a previous-token head and an induction head match the query to its earlier occurrence and copy the following label. As each component is known, damage from recursive training can be traced to specific heads and connections rather than inferred from accuracy.

This study therefore trains transformers on this task over generations whose training data are partly or wholly labelled by the previous generation's model, comparing the replacement of 50–100% of the real data with accumulation, for a base model and an extended model that also generates part of each sequence (Figure 1). The circuit of every generation is traced and tested causally. A well-trained model reproduces its own labels almost exactly, so collapse appears only once sampling noise introduces errors. The circuit's margin for the correct label then halves while accuracy is unchanged, and this erosion appears to set the error rate of the next generation's data, whereas accumulating real data keeps accuracy at 0.99 and the circuit compact, although its margin still declines.

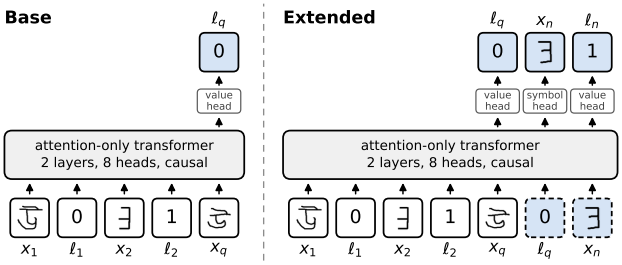


Figure 1: Base (left) and extended (right) models. The base model predicts the query's label ℓ_q from two symbol–label pairs and query x_q. The extended model additionally predicts the next symbol x_n and its label ℓ_n. Shaded boxes indicate model outputs, while dashed inputs represent true tokens during training and generated tokens during recursive training.

Background

Induction heads The previous-token head copies each symbol into the label that follows it, and the induction head then matches the query against these copies through its key input and copies the matching label to the output (Singh et al., 2024). Several induction heads can form and contribute additively (Singh et al., 2024), so the circuit can spread across heads rather than rely on one, and interventions on their formation change ICL ability with them (Chen et al., 2026).

Model collapse Shumailov et al. (2024) identify sampling error as the primary cause of collapse: finite samples rarely reproduce low-probability events, so the tails of the distribution vanish first, and each model's errors compound in the data of the next. In fine-tuned OPT-125m, preserving 10% of the original data led to only minor degradation.

Accumulating data Gerstgrasser et al. (2024) found that replacing real data causes test error to grow with every iteration, whereas accumulating synthetic data alongside it keeps the error bounded, across language, diffusion, and autoencoder models. This study tests whether the same contrast holds for the induction circuit.

Methods

This study varies the share of real data in the training pool and measures its effect on the induction circuit. It uses the setup of Singh et al. (2024), whose circuit is known to form: each sequence holds two exemplar–label pairs and a query, with Omniglot exemplars encoded by a frozen ResNet18 and five labels reassigned per sequence. The model, a causal two-layer attention-only transformer with eight heads per layer, trains on 50 classes and is tested on 100 unseen classes, so that accuracy reflects ICL alone.

Base and extended models The base model predicts the query's label. The extended model also predicts the next symbol, through a separate symbol head, and that symbol's label (Figure 1), so that three tokens per sequence become synthetic rather than one.

Next-symbol selection By default, the next symbol is a context symbol chosen uniformly, a conservative baseline in which every target remains solvable by the induction circuit. A second condition draws it from all training classes, so that the model must learn an unconditional distribution. Generated tokens are sampled from the softmax at temperature T, the source of error described above; argmax removes it and serves as an ablation.

Real-to-synthetic ratio Each run trains on a fixed pool of $N = 20{,}000$ sequences over six generations, the first on real data. The previous model then regenerates the targets of 0%, 50%, 90%, or 100% of the pool from their real prompts, where 0% is a real-only control and 90% matches the 10% of real data kept by Shumailov et al. (2024). Accumulation instead keeps every item and adds N synthetic sequences per generation (Gerstgrasser et al., 2024), matching 50% replacement at generation 1, so that keeping earlier data can be separated from the real share itself. A fixed initialisation and training length ensure that conditions differ only in their pool.

Forcing synthetic errors At $T = 1$, the tuned model regenerates its labels almost perfectly, with 0.01% wrong after 20 generations, so every ratio yields an essentially real pool. Each regenerating condition is therefore also run at $T = 3$ and $T = 5$, which introduce 0.5% and 6.5% wrong labels, so that the ratio acts at a known error rate.

Tuning As collapse can only alter a circuit that has formed, the learning rate and training length are tuned per model on generation 0, using 1,473 validation classes so that the test classes remain unseen. This gives $\eta = 10^{-4}$ for the base model and 3×10^{-4} for the extended model, both for one million sequences.

Burstiness and class distribution Two sweeps at $T = 1$ test whether properties of the data admit errors without forcing. Burstiness, $B \in \{1, 2, 4\}$, sets how many of eight context items share the query's class: a low B leaves the induction head a single match among many items, which aims to elicit a less confident model whose own errors enter the pool, while a high B provides redundant matches. Zipfian class frequencies, $\alpha \in \{0.5, 1, 1.5\}$, create a long tail of rare symbols.

Circuit analysis Each generation's model is analysed on the same real sequences, so that changes reflect the model rather than the data. The logit difference measures how strongly the circuit separates the correct label, the number of connections Automatic Circuit DisCovery (ACDC) (Conmy et al., 2023) keeps at a fixed $\tau = 0.001$ measures whether it stays sparse, and the induction score measures whether the induction head still attends to the match. Patching the previous-token head's output alone into the induction head tests whether the circuit still routes through it, and test accuracy is reported alongside, since it can stay high while the logit difference falls.

Results

Table 1 summarises every condition at its final generation, and the paragraphs below follow the questions in turn.

No errors, no collapse At $T = 1$, the tuned base model regenerates its labels almost exactly, and the 50%, 90%, and 100% replacement runs produce identical models, as does argmax sampling relative to the real-only control. Accuracy therefore stays at 0.99 under every ratio. The circuit nonetheless drifts over 20 generations of full replacement: its margin falls from 19.9 to 12.5 and its ACDC circuit grows from 17 to 35 edges (Figure 2a–b), whereas the real-only margin varies only between 16.9 and 20.0 across generations. This suggests that the circuit changes before accuracy even when almost no errors enter the data.

Table 1: Final generation of each condition. Without forced errors, accuracy holds everywhere except the least bursty context, while the margin falls and the circuit grows. With errors forced at $T = 3$, accumulation keeps accuracy and the induction score near the real-only level for five generations, but declines slowly over 20, whereas each replacement level settles at an accuracy set by its real share. The real-only row never samples and serves as the control at every temperature. †Margin below 1, where the circuit size is less reliable. Induction scores are omitted for eight-item contexts, where they vary between real-only retrains.

		Real	Label	Base model				Extended model	
Condition	Gen.	share	error	Acc.	Margin	Edges	Ind.	Margin	KL
No forced errors ($T = 1$)									
Real only	5	100%	0.00	0.99	20.0	19	0.97	16.0	0.00
100% replaced	19	0%	0.01	0.99	12.5	35	0.87	13.8	0.03
Accumulate	19	5%	0.00	1.00	10.4	22	0.91	6.3	0.01
100% replaced, $B = 1$	5	0%	0.06	0.75	11.0	49	–	7.1	0.12
100% replaced, $B = 4$	5	0%	0.00	0.99	13.2	21	–	12.5	0.02
Forced errors ($T = 3$)									
50% replaced	5	50%	0.30	0.80	3.0	198	0.84	4.6	0.00
90% replaced	5	10%	0.66	0.38	0.6	218†	0.27	0.2	0.00
100% replaced	5	0%	0.75	0.25	0.1	209†	0.05	0.0	0.00
Accumulate	5	17%	0.31	0.99	2.1	23	0.95	3.1	0.00
Forced errors over 20 generations ($T = 3$, base model only)									
50% replaced	19	50%	0.33	0.78	2.7	190	0.85	–	–
90% replaced	19	10%	0.69	0.31	0.4	204†	0.16	–	–
100% replaced	19	0%	0.79	0.19	0.1	210†	0.04	–	–
Accumulate	19	5%	0.59	0.93	0.8	12†	0.80	–	–

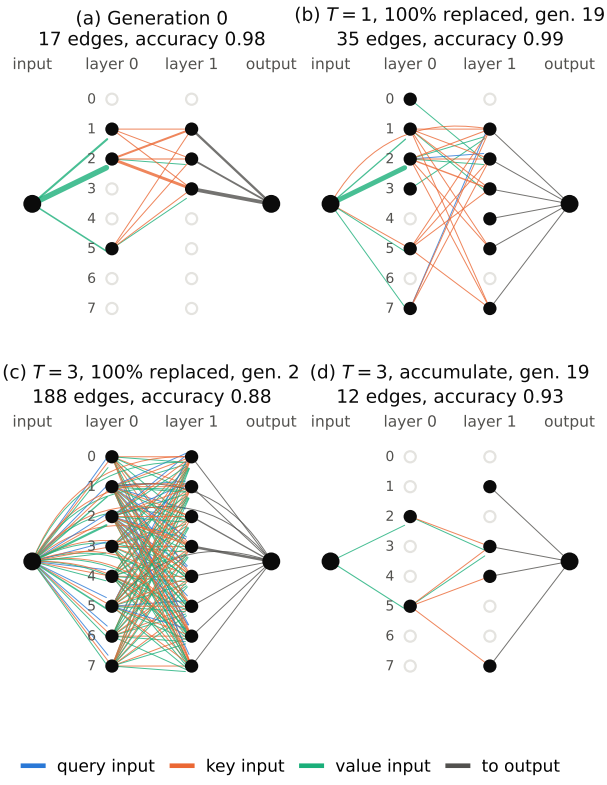


Figure 2: ACDC circuits of the base model on a fixed layout of the input, the eight heads of each layer, and the output, with filled nodes marking heads in the circuit and edge width growing with its effect on the output. At generation 0 (a), the circuit routes the input through the previous-token head A0.2 into the key input of the induction head A1.3. Twenty error-free generations of full replacement (b) double its size while accuracy holds, and

The circuit fails before accuracy When sampling at $T = 3$ introduces errors, the circuit degrades in a consistent order under full replacement. The margin halves by generation 1, from 19.9 to 10.8, while accuracy holds at 0.99. By generation 2, the circuit has grown from 17 to 188 edges, spreading across every head and into query inputs that the original circuit did not use (Figure 2c), while accuracy is still 0.88. Accuracy then falls to 0.25 by generation 5, and the induction score falls last, from 0.75 at generation 3 to 0.05, with the previous-token head holding until generation 4. Accuracy then settles near chance, at 0.19 by generation 19, once 79% of the pool's labels are wrong, close to the 80% expected from five labels assigned at random. This order is likely because noisy labels make the label only partly predictable from the context, so that the model lowers its confidence and spreads the computation across more connections, consistent with the additive contribution of several heads reported by Singh et al. (2024). The extended model follows the same order, with its circuit growing from 46 to 179 edges while accuracy is still 0.88.

The margin sets the next generation's errors If the correct label leads each of the four wrong labels by the margin D_g of generation g's model, sampling at temperature T produces wrong labels at a rate of

$$\hat{\varepsilon}_{g+1} = \frac{4\,e^{-D_g/T}}{1+4\,e^{-D_g/T}}. \tag{1}$$

The observed error rate of each generation's new samples closely matches Equation 1 across both models, both temperatures, and every condition, with a correlation of 0.994 and a

mean absolute error of 0.021. At $T = 3$, the generation 0 margin of 19.9 predicts 0.5% wrong labels, while the margin of 5.3 at generation 2 predicts 41% against an observed 44%. The eroding margin is therefore not only a symptom of collapse: it determines how many errors the next generation inherits, which explains why errors compound once they begin.

Retaining real data Over 20 generations at $T = 3$, each replacement level settles at an accuracy set by its real share, at 0.78 with 50% real data, 0.31 with 10%, and 0.19 with none (Figure 3). With 50% real data, the pool's label error levels off at 33% and the margin at about 2.7, so that retaining half the real data bounds the collapse but does not prevent the circuit from spreading to 190 edges. Accumulation holds accuracy at 0.97 or above for 15 generations, even though its real share falls to 6% by then, and at generation 19 it still reaches 0.93 with only 5% real data, against 0.78 for 50% replacement. As both conditions hold 50% real data at generation 1, keeping earlier data appears to matter more than the real share itself, likely because each real prompt keeps its correct label while errors scatter across the four wrong labels. However, accumulation delays collapse rather than preventing it: its margin falls steadily from 19.9 to 0.77, the share of wrong labels in its pool rises to 59%, and accuracy begins to decline once the margin drops below 1 at generation 16. Its circuit contracts to 12 edges rather than spreading (Figure 2d), although at a margin below 1 this count is less reliable. This contrasts with Gerstgrasser et al. (2024), who found that accumulation keeps the error bounded, and suggests that the margin of the circuit erodes even when its behaviour is protected.

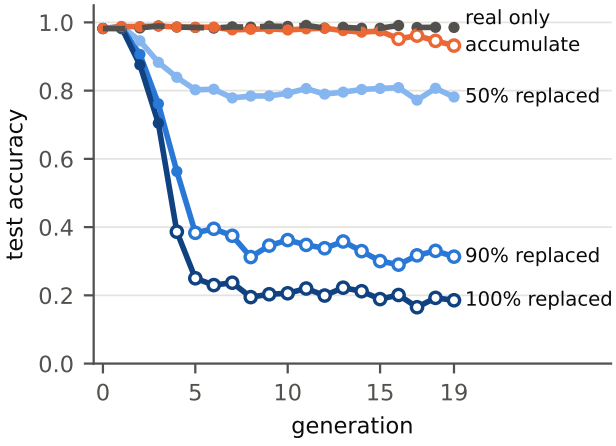


Figure 3: Test accuracy of the base model per generation at $T = 3$ for each condition over 20 generations. Each replacement level settles at an accuracy set by its real share, while accumulation stays with the real-only control for 15 generations before declining slowly. Hollow markers indicate a margin below 1.

Symbol distribution and data properties The extended model's symbol distribution does not narrow: its divergence from generation 0 stays at or below 0.02, even when 96% of its next symbols fall outside the context at $T = 3$, so that it collapses through wrong copying rather than a lost tail. Burstiness shows the clearest effect of the data without forced errors. Under full replacement at $T = 1$, the extended model loses 57%, 39%, and 14% of its margin relative to real-only retraining as B rises from 1 to 4, and its symbol divergence falls from 0.12 to 0.02, while only the base model at $B = 1$ admits enough errors to lose accuracy, from 0.83 to 0.75. Redundant matches in the context therefore appear to protect the circuit, likely because they keep the model confident enough that its own samples remain correct. The Zipfian sweeps, in contrast, leave accuracy and the tail mass of the symbol distribution unchanged, likely because the next symbol is copied from real prompts, which preserve rare symbols by construction.

Discussion

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Computations were performed using High Performance Computing infrastructure provided by the Mathematical Sciences Support Unit at the University of the Witwatersrand.

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